

NiFi ConvertRecord to transform nested json to csv

Asked 1 month ago Active 1 month ago Viewed 72 times

I have to convert nested json to csv: Input Json:

0

```
[
  {
    "phone": "657-676-6697",
    "city": "London",
    "rc": [],
    "rcName": "Andy"
  },
  {
    "phone": "",
    "city": "Paris",
    "rc": [
      {
        "email": null,
        "firstName": "Mary",
        "lastName": "Kim",
        "title": "Teacher"
      },
      {
        "email": "anna@gmail.com",
        "firstName": "Anna",
        "lastName": "Frank",
        "title": "Student"
      }
    ],
    "rcName": ""
  },
  {
    "phone": "",
    "city": "",
    "rc": [ {
      "email": "pan@tt",
      "firstName": "Pan",
      "lastName": "Kurt",
      "title": "Teacher"
    } ],
    "rcName": "Pan1"
  },
  {
    "phone": "",
    "city": "Oklahoma",
    "rc": [
      {
        "email": null,
        "firstName": "Pearl",
        "lastName": "Knox",
        "title": "Receptionist"
      },
      {
        "email": "Jim@gmail.com",
        "firstName": "Jim",
        "lastName": "Hopkins",
        "title": "TA"
      }
    ],
    "rcName": ""
  }
]
```

Expected Output csv:

```
city,lastName,title
London,
Paris, Kim, Teacher
Paris, Frank, Student
, Kurt, Teacher
Oklahoma, Knox, Receptionist
Oklahome, Hopkins, TA
```

Converting to csv for phone, city and rcName using ConvertRecord is working perfectly fine for me. But it gives empty values for elements in nested json. There were recommendations at link <https://www.titanwolf.org/Network/q/2e265d53-43b9-42a4-94ad-6cbcc44cf395/y> to transform the input JSON into "rows" using JoltTransformJSON. So, I am using following spec,

```
[
  {
    "operation": "shift",
    "spec": {
      "*": {
        "rc": {
          "*": {
            "@(2,city)": "[&1].city",
            "@(2,phone)": "[&1].phone",
            "*": "[&1].&"
          }
        }
      }
    }
  }
]
```

But, this JoltTransform is giving all the values in array and ConvertRecord is not able to parse it. Jolt Transform output:

```
[{
  "city": ["Paris", "", "Oklahoma"],
  "phone": ["", "", ""],
  "email": ["pan@tt", null],
  "firstName": ["Mary", "Pan", "Pearl"],
  "lastName": ["Kim", "Kurt", "Knox"],
  "title": ["Teacher", "Teacher", "Receptionist"]
}, {
  "city": ["Paris", "Oklahoma"],
  "phone": ["", ""],
  "email": ["anna@gmail.com", "Jim@gmail.com"],
  "firstName": ["Anna", "Jim"],
  "lastName": ["Frank", "Hopkins"],
  "title": ["Student", "TA"]
}]
```

Jolt Transform Demo Using v0.1.1

Here you can experiment with the stock Jolt Transforms without having to download and run the Java code.

Json InputJSON Validate

```
1 [{
2   {
3     "phone": "657-676-6697",
4     "city": "London",
5     "rc": [],
6     "rcName": "Andy"
7   },
8   {
9     "phone": "",
10    "city": "Paris",
11    "rc": [
12      {
13        "email": null,
14        "firstName": "Mary",
15        "lastName": "Kim",
16        "title": "Teacher"
17      },
18      {
19        "email": "anna@gmail.com",
20        "firstName": "Anna",
21        "lastName": "Frank",
22        "title": "Student"
23      }
24    ],
25    "rcName": ""
26  },
27  {
28    "phone": "",
29    "city": "",
30    "rc": [
31      {
32        "email": "pan@ttt",
33        "firstName": "Pan",
34        "lastName": "Kurt",
35        "title": "Teacher"
36      }
37    ]
38  }
39 ]
```

Jolt SpecJSON Validate

```
1 [{
2   {
3     "operation": "shift",
4     "spec": {
5       "*": {
6         "rc": {
7           "*": {
8             "@(2,city)": "[&1].city",
9             "@(2,phone)": "[&1].phone",
10            "*": "[&1].&"
11          }
12        }
13      }
14    }
15  }
16 ]
```

Output / ErrorsTransformSort Output?

```
1 [{
2   "city": [ "Paris", "", "Oklahoma" ],
3   "phone": [ "", "", "" ],
4   "email": [ "pan@ttt", null ],
5   "firstName": [ "Mary", "Pan", "Pearl" ],
6   "lastName": [ "Kim", "Kurt", "Knox" ],
7   "title": [ "Teacher", "Teacher", "Receptionist" ]
8 }, {
9   "city": [ "Paris", "Oklahoma" ],
10  "phone": [ "", "" ],
11  "email": [ "anna@gmail.com", "jim@gmail.com" ],
12  "firstName": [ "Anna", "Jim" ],
13  "lastName": [ "Frank", "Hopkins" ],
14  "title": [ "Student", "TA" ]
15 }]
```

I have used ConvertRecord processor with following configuration: RecordReader: JsonPathReader RecordWriter: CSVRecordSetWriter

json apache-nifi jolt

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edited Sep 9 at 17:13

asked Sep 9 at 3:32

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- 1 Please share output of jolt and specify what exact issue do you have. Using your jolt spec and your example i have got [{ "city" : "Paris", "email" : null, "firstName" : "Mary", "lastName" : "Kim", "title" : "Teacher" }, { "city" : "Paris", "email" : "anna@gmail.com", "firstName" : "Anna", "lastName" : "Frank", "title" : "Student" }] which looks ok to me as an input to ConvertRecord – YuriR Sep 9 at 10:21

@YuriR I have updated my question with JoltTransform output. For more data json is coming in array – likeGreen Sep 9 at 13:53

this is not what i get on jolt-demo.appspot.com. Please try there. Probably nifi has a bug or you copied wrong Jolt Spec to your question :-> – YuriR Sep 9 at 14:46

@YuriR I am getting same output from both Nifi and link shared by you. Screenshot attached in question – likeGreen Sep 9 at 15:51

can reproduce now. will update if i have progress. – YuriR Sep 9 at 16:10

1 Answer

Active Oldest Votes



I used below JoltTransform spec which solved my issue

1



```
[
{
  "operation": "shift",
  "spec": {
    "*": {
```



```
"rc": {
  "*": {
    "@(2,city)": "[&3].[&1].city",
    "@(2,phone)": "[&3].[&1].phone",
    "*": "[&3].[&1].&"
  }
}
},
{
  "operation": "shift",
  "spec": {
    "*": {
      "*": "[]"
    }
  }
}
]
```

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[edited Sep 10 at 7:18](#)



[Barbaros Özhan](#)

41.3k 9 22 48

[answered Sep 10 at 3:48](#)



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